

6 (a) Explain how human lungs are adapted for rapid gas exchange.

(3)

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(b) Cystic fibrosis is an inherited condition which reduces gas exchange.

The most frequently observed CFTR allele associated with cystic fibrosis carries the $\Delta F508$ mutation.

(i) State what is meant by the term **allele**.

(1)

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- (ii) The CFTR protein coded for by this mutation has one missing amino acid compared to the functioning protein.

Explain how this mutation results in a non-functioning CFTR protein.

(4)

- (iii) Explain why people with cystic fibrosis may develop lung infections.

(4)

(Total for Question 6 = 12 marks)